

CS1231S: Discrete Structures
Tutorial #10: Counting and Probability II and Graphs I
(Week 12: 8 – 12 April 2024)

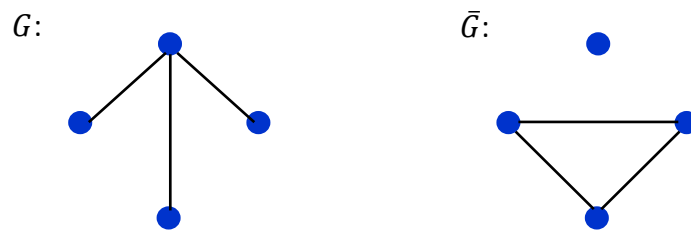
I. Discussion Questions

- D1. Suppose a random sample of 2 lightbulbs is selected from a group of 8 bulbs in which 3 are defective, what is the expected value of the number of defective bulbs in the sample? Let X represent of the number of defective bulbs that occur on a given trial, where $X = 0, 1, 2$. Find $E[X]$.
- D2. How many **injective functions** are there from a set A with m elements to a set B with n elements, where $m \leq n$?
- D3. How many **surjective functions** are there from a 5-element set A to a 3-element set B ?

II. Definitions

Definition 1. If G is a simple graph, the *complement* of G , denoted $\bar{G}$, is obtained as follows: the vertex set of $\bar{G}$ is identical to the vertex set of G . However, two distinct vertices v and w of $\bar{G}$ are connected by an edge if and only if v and w are not connected by an edge in G .

The figure below shows a graph G and its complement $\bar{G}$.



A graph G and its complement $\bar{G}$.

Definition 2. A *self-complementary* graph is isomorphic with its complement.

Definition 3. A simple circuit (cycle) of length three is called a *triangle*.

III. Tutorial Questions

- Find the term independent of x in the expansion of $\left(2x^2 + \frac{1}{x}\right)^9$.
- Let's revisit Tutorial #9 Question 4:

Given n boxes numbered 1 to n , each box is to be filled with either a white ball or a blue ball such that at least one box contains a white ball and boxes containing white balls must be consecutively numbered. What is the total number of ways this can be done?

Last week, the answer given was

For k ($1 \leq k \leq n$) consecutively numbered boxes that contain white balls, there are $n - k + 1$ ways. Therefore, total number of ways is $\sum_{k=1}^n (n - k + 1) = \sum_{k=1}^n k = \frac{n(n+1)}{2}$.

Now, let's use another approach to solve this problem. Draw crosses on the side of the boxes as shown below. How do you use these crosses?



- [AY2020/21 Semester 2 Exam Question]
On a die there are 6 numbers. We call 4, 5, 6 the big numbers and 1, 2, 3 the small numbers. Given a loaded die in which the probability of rolling any fixed big number is twice the probability of rolling any fixed small number, answer the following questions.
 - What is the probability of rolling a 6? Write your answer as a single fraction.
 - If two such loaded dice are rolled, what is the expected value of the maximum of the two dice? Write your answer as a single fraction.
- An urn contains five balls numbered 1, 2, 2, 8 and 8. If a person selects a set of two balls at random, what is the expected value of the sum of the numbers on the balls?

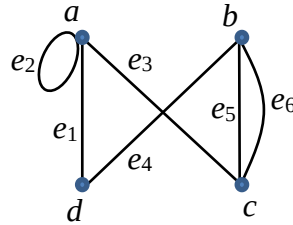
- [AY2020/21 Semester 1 Exam Question]
In 1995, Mars® introduced the blue M&M's to replace the tan M&M's. Before that, the colour mix in a bag of plain M&M's was 30% brown, 20% yellow, 20% red, 10% green, 10% orange and 10% tan. After the replacement, the colour mix was 24% blue, 16% green, 20% orange, 14% yellow, 13% red and 13% brown.



Your friend Aiken, being a big fan of M&M's, has two bags of plain M&M's, one from 1994 and the other from 1996. He won't tell you which is which, but he gives you one M&M from each bag. One is yellow and one is green. What is the probability that the **yellow M&M came from the 1994 bag**, assuming the M&M's are picked randomly.

You may express your answer as a fraction or a decimal number correct to 3 significant digits

6. Given the graph shown below:



- Give the adjacency matrix A for the graph, with vertices in the order a, b, c, d .
- Compute A^0 , A^2 and A^3 .
- How many walks of length 2 are there from a to b ? From c to itself? List out all the walks.
- How many walks of length 3 are there from a to c ? List out all the walks.

7. A lady hosted a party of n ($n \geq 2$) people (including herself). At the party, various friends met and some of them shook hands with each other. The thoughtful host made sure that she shook hands with everyone at the party.

Prove that there are at least two people who have shaken hands the same number of times.

8. [AY2015/16 Semester 1 Exam Question]

Let $A = \{1, 2, 3, 4\}$. Since each element of $P(A \times A)$ is a subset of $A \times A$, it is a binary relation on A . ($P(S)$ denotes the powerset of S .)

Assuming each relation in $P(A \times A)$ is equally likely to be chosen, what is the probability that a randomly chosen relation is (a) reflexive? (b) symmetric?

Can you generalize your answer to any set A with n elements?

9. [AY2021/22 Semester 1 Exam]

A set of vertices, D , in an undirected simple graph is said to be a **dominating set** if every vertex not in D is adjacent to at least one vertex in D . A **minimal dominating set** is a dominating set such that none of its proper subsets are dominating sets.

- Draw two non-isomorphic simple graphs with four vertices that have minimal dominating sets of size one. Highlight the vertices in the minimal dominating set in your graphs.
- Draw two non-isomorphic simple graphs with four vertices that have minimal dominating sets of size three. Highlight the vertices in the minimal dominating set in your graphs.

10. Let G be any simple graph with 6 vertices. Prove that G or its complementary graph $\bar{G}$ contains a triangle. (This is similar to the problem of showing that in any group of 6 people, there must be either 3 mutual friends or 3 mutual strangers. But for this question, please use the graph formulation.)

The following definitions are used in Question 11.

A relation $<$ on a set A is said to be **irreflexive**, if and only if, $\forall a \in A, (a \not< a)$.

Alternate definition of anti-symmetry: $\forall x, y, (x \neq y) \Rightarrow ((x, y) \in R) \Rightarrow ((y, x) \notin R)$.

A relation is a **strict partial order** if and only if it is irreflexive, antisymmetric and transitive.

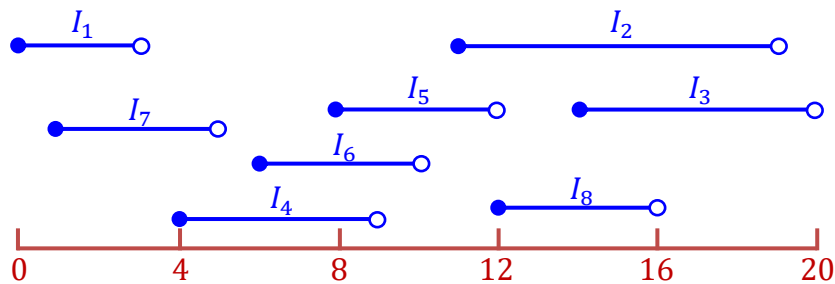
Let $\leq$ be a strict partial order on a set A . A subset C of A is called **chain** if and only if each pair of *distinct* elements in C is comparable, that is, $\forall a, b \in C (a \neq b) \Rightarrow (a \leq b \vee b \leq a)$.

A **maximal chain** is a chain M such that $t \notin M \Rightarrow M \cup \{t\}$ is not a chain.

11. You are given a set of n jobs $J = \{J_1, J_2, J_3, \dots, J_n\}$. Each job J_k is represented by the interval $I_k = [s_k, e_k)$ where s_k is the *start time* and e_k (where $s_k < e_k$) is the *end time* of the job, for $k = 1, 2, \dots, n$. An instance of this problem with $n = 8$ is shown below.

Instance: $n = 8$, and $\{J_1, J_2, J_3, \dots, J_8\}$, and

$$\begin{array}{llll} I_1 = [0, 3), & I_2 = [11, 19), & I_3 = [14, 20), & I_4 = [4, 9), \\ I_5 = [8, 12), & I_6 = [6, 10), & I_7 = [1, 5), & I_8 = [12, 16), \end{array}$$



Define a relation $<$ on J by: $(J_x < J_y) \Leftrightarrow (e_x \leq s_y)$

Namely, $J_x < J_y$ means that the job J_x is *to the left of* the job J_y .

- Show that the relation $<$ is a strict partial order.
- Draw the graph $G = (J, <)$ of the relation $<$ for this instance given above.

You want to assign all the jobs to workers so as to *minimize* the number of workers deployed. Each worker can only work on one job at a time. Each worker can do any number of jobs as long as the jobs do not overlap in time. (You can assume that a worker can (if necessary) start on a new job immediately after finishing a previous job. In the instance above, jobs J_5 and J_8 can be assigned to the same worker, if necessary.) This problem can be solved in parts (c) and (d) as follows:

- Give a maximal chain C in the graph G . Argue that the jobs in C can be assigned to a single worker. (Refer to the definition of *maximal chain* in midterm test.)
- Now, partition the vertices in G into a *minimum* number of vertex-disjoint chains. For each chain, we assign a worker to do all the jobs in that chain. What is the minimum number of workers needed? Show the jobs assigned to each worker (W1, W2, etc.).